

Elapsed time

Synthetic integer workload | One machine | Apple M3 MacBook Air, 16 GB RAM
5 independent repeats per cell | 270 measured runs | Same source data and exact result checks

● Python loops ■ pandas ▲ SQLite Small marks: individual runs Large marks: median Bars: observed min-max



Bars show the observed minimum and maximum, not confidence intervals. Dataset sizes are equally spaced categories.
Time y-axis limits differ between cold and warm rows; they are shared across operations within each mode.
Cold time includes loading, backend preparation and query output. Warm time includes only the query and its complete output.
RSS is the peak of the whole worker through query completion: imports, source data, backend setup and warm priming included.
Process-cold is not disk-cold; OS file cache was not cleared. One machine, one synthetic schema. No energy or CO2 measurement.